

Course Outline

Robust and Data-driven Simulation

Probability Theory

Introduction to Simulation

Input probability distributions

Random number generation

Monte Carlo Estimates and Accuracy

Trajectories of Markov Chains

Discrete Event Simulation

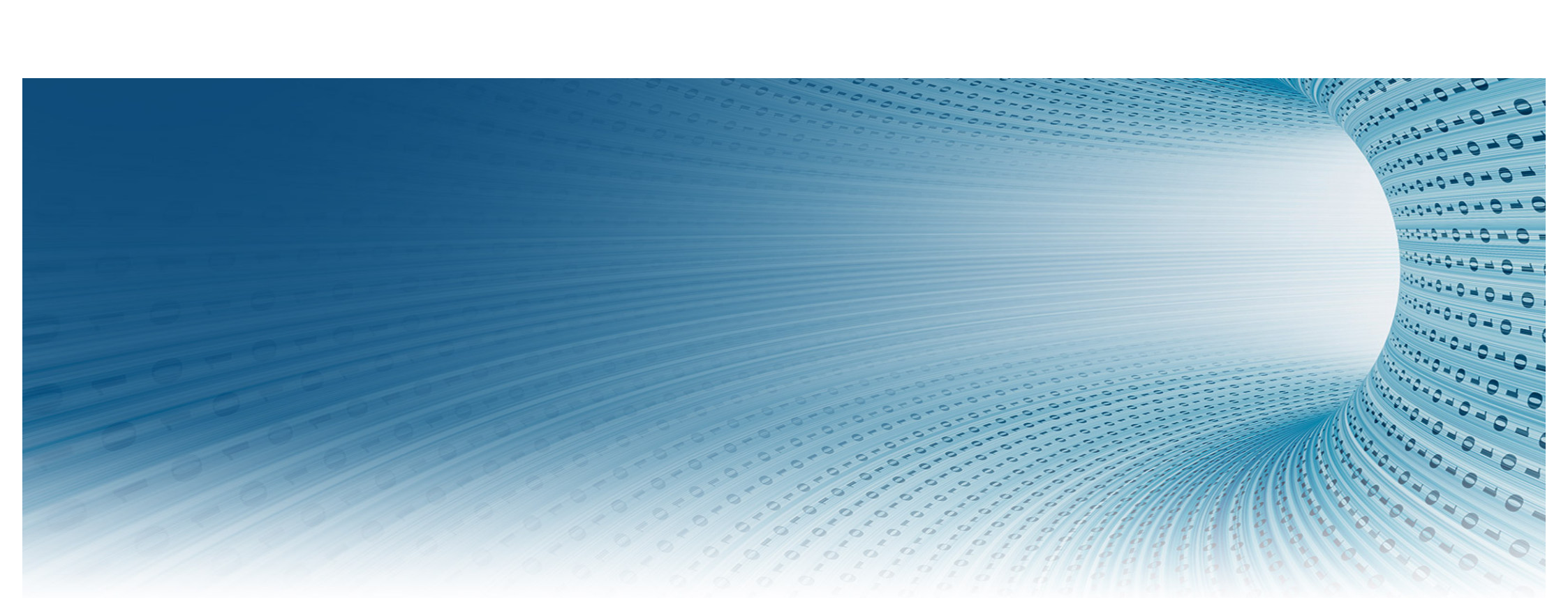
Experimental Design and Optimisation

Robust and Data-driven Optimisation

Chance-constrained optimisation

Stochastic optimisation

Robust optimisation



Monte Carlo Estimates and Accuracy

Theory and formulation

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Outline

- Introduction
- Sampling
- Monte Carlo Integration
- Variance-reducing techniques

Introduction

- Simulated trials leads to an outcome or a single result
 - E.g. A head or tail resulting from a tossing experiment
 - Each execution produces *one output or result*, i.e. *a scalar or a vector*, which is the result of a function of the random numbers used
 - Running the simulation again will produce a different result
 - Questions
 - How many times do we need to perform the simulation?
 - What do we choose as final answer?
 - How do we measure the accuracy?
- ⇒ Central to these questions is the concept of **sampling**

Sampling

- Example
 - The city council of Ghent wants to pass new laws concerning owning pets but they have decided to first conduct a survey to determine the number of residents that own a pet. Ghent is a city of approximately 250,000 inhabitants.
 - In order to conduct the survey the following questions arise
 - How many of the town's population they need to contact and interview?
 - How to select the individuals?
 - How to infer the number of citizens in Ghent that own a pet based on the information they collect?

Sampling

- Terminology

- The set of individuals is called the *sample set*
- The size of the sample set is the *sample size*
- The list of all possible individuals, constitutes the *population*

- Two types of errors

- Sampling bias

- This error arises when the sample set does not represent the population
- No matter how large the sample size, this sample bias would not be removed
- If the set of individuals completely is chosen at random (i.e. each individual has an equal chance of being selected), the sample is unbiased

- Sampling variability

- The unrepresentative selection occurs by accident
- This bias is more likely to occur when the sample size is small
- Unlike sample bias, sample variability can be reduced by choosing a larger sample size

Sampling

- Point estimators

- Definition: A point estimator returns the estimated value of some parameter

- Point estimators of a sample

- The sample mean $\bar{X}$ is an estimator of the overall population mean μ

- The mean value for a sample set of size n is equal to $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ with X_i is the i^{th} individual in the sample set

- The expectation is equal to the overall population mean μ , i.e. $E[\bar{X}] = \frac{1}{n} \sum_{i=1}^n E[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \mu$

- The estimator of a parameter is **unbiased** when its expectation is equal to the parameter

- The estimator of a parameter is **asymptotically unbiased** when it tends to the parameter in the limit as n goes to infinity

- The sample variance S^2 is an estimator of the overall population variance σ^2

- S^2 is an unbiased estimator if $E[S^2] = \sigma^2$ or $S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}$

Sampling

- Point estimators of the sampling distribution

- The mean of the sampling means $\bar{X}$ is the overall population mean μ

- The variance of $\bar{X}$ given the overall population variance σ^2

- The sample variance for a sample set is equal to $\text{Var}[\bar{X}] = \text{Var}\left[\frac{X_1 + X_2 + \dots + X_n}{n}\right] = \frac{1}{n^2} \text{Var}[X_1 + X_2 + \dots + X_n] = \frac{\sigma^2}{n}$
- The estimator whose value goes to zero with increased sample size n is a **consistent** estimator, i.e. the accuracy of $\bar{X}$ as an estimator of the population mean μ increases as the sample size grows, which is independent of the population size. In other words $\lim_{n \rightarrow \infty} \bar{X} = \mu$ with probability 1
- The variance of a random variable $\bar{X}$ having a mean value μ is defined as $\text{Var}[\bar{X}] = E[(\bar{X} - \mu)^2] = \frac{\sigma^2}{n}$

⇒ A good estimator is both unbiased and consistent

Sampling

- Interval estimators/Confidence intervals

- Interval estimators return information how the point estimator differs from the true value
- Interval estimators find the probability ρ that the true value ξ lies in some interval $(\alpha - \epsilon_1, \alpha + \epsilon_2)$ surrounding the estimator value α , i.e.

$$\text{Prob}\{\alpha - \epsilon_1 < \xi < \alpha + \epsilon_2\} = \rho$$

- The interval $(\alpha - \epsilon_1, \alpha + \epsilon_2)$ is a $100 \times \rho$ percent confidence interval

Sampling

- Interval estimators/Confidence intervals

- The confidence interval for the sample mean (or any other statistic)

- Chebychev inequality

$$\text{Prob}\{|\bar{X} - \mu| \geq \epsilon\} \leq \frac{\sigma^2/n}{\epsilon^2} \quad \text{or} \quad \text{Prob}\{-\epsilon < \bar{X} - \mu < \epsilon\} = \text{Prob}\{|\bar{X} - \mu| < \epsilon\} \geq 1 - \frac{\sigma^2}{n\epsilon^2}$$

This probability is close to 1 if the sample size n is large

- Suppose the population from which the sample set is drawn normally distributed $\sim N(\mu, \sigma^2)$. Via the Central limit theorem we know

- The sample mean $\bar{X}$ is normally distributed $\sim N(\mu, \frac{\sigma^2}{n})$
 - The random variable $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$ or

$$\text{Prob}\{|Z| < z\} = \text{Prob}\left\{\frac{|\bar{X} - \mu|}{\sigma/\sqrt{n}} < z\right\} = \text{Prob}\left\{|\bar{X} - \mu| < \frac{z\sigma}{\sqrt{n}}\right\} = \text{Prob}\left\{\bar{X} - \frac{z\sigma}{\sqrt{n}} < \mu < \bar{X} + \frac{z\sigma}{\sqrt{n}}\right\} = \rho$$

- (100- ζ)-confidence interval for the population mean: $[\bar{X} - \frac{z_{\zeta/2}\sigma}{\sqrt{n}}, \bar{X} + \frac{z_{\zeta/2}\sigma}{\sqrt{n}}]$

$1-\zeta$	0.8	0.9	0.95	0.98	0.99	0.998
$z_{\zeta/2}$	1.282	1.645	1.96	2.326	2.576	3.09

Sampling

- Interval estimators/Confidence intervals

- Central limit theorem

- If the population distribution is not normal, then the distribution of $\bar{X}$ is not normal either. However, if the underlying distribution has a mean μ and variance σ^2 , then $\bar{X}$ approaches the normal distribution with mean μ and variance σ^2/n or

$$\text{Prob}\{|\bar{X} - \mu| \geq \epsilon\} \approx \text{Prob}\left\{|Z| > \frac{\epsilon\sqrt{n}}{\sigma}\right\} = 2[1 - \Phi(\frac{\epsilon\sqrt{n}}{\sigma})]$$

where $\Phi(x)$ is the cumulative standard normal distribution function. If $\frac{\epsilon\sqrt{n}}{\sigma} = 1.645$ we find

$$\text{Prob}\{|\bar{X} - \mu| \geq \epsilon\} \approx \text{Prob}\{|Z| > 1.645\} = 2[1 - \Phi(1.645)] = 0.1 \quad \text{since } \Phi(1.645) = 0.95$$

- Applicability
 - Valid for different types of sample statistics such as the mean, the variance, the sample proportion, etc
 - We assume that the population size is much larger than the sample size, then the sampling distribution has roughly the same standard error with or without replacement
 - The sample size n should be large enough (at least 30 or more)
 - Requirements for accuracy: The more closely the sampling distribution needs to resemble a normal distribution, the more sample points will be required
 - The shape of the underlying population: The more closely the original population resembles a normal distribution, the fewer sample points will be required

Sampling

- Interval estimators/Confidence intervals

- The half-width $\frac{z\sigma}{\sqrt{n}}$ is a measure of the accuracy of the sample mean

- The error decreases extremely slowly in terms of the number of samples

- In order to double the accuracy, a fourfold increase in the sample size is required

- In order to increase the accuracy by a factor 10, 100 times more samples are required

- When is the standard deviation σ of the population is not known, we should use S^2

- Remark: The sample size should be at least 30 to compute the confidence interval

- For small sample sizes the random variable $T = \frac{|\bar{X} - \mu|}{S/\sqrt{n}}$ has a Student T-distribution

with $n-1$ degrees of freedom (e.g. $n = 6$)

$1-\zeta$	0.8	0.9	0.95	0.98	0.99	0.998
$z_{\zeta/2}$	1.476	2.015	2.571	3.365	4.032	5.893

Sampling

- Interval estimators/Confidence intervals: Example I

- Consider 6 independent random samples X_i from a normal distribution with mean $\mu = 5$ and variance $\sigma^2 = 0.3$

4.789; 5.021; 4.856; 4.691; 5.255; 4.703

The sample mean $\bar{X} = \frac{1}{6} \sum_{i=1}^6 X_i = 4.8858$

- When the variance is known $\sigma^2 = 0.3$

95% confidence interval ($z_{\zeta/2} = 1.96$): $[\bar{X} - \frac{z_{\zeta/2}\sigma}{\sqrt{n}}, \bar{X} + \frac{z_{\zeta/2}\sigma}{\sqrt{n}}] = [4.4476, 5.3421]$

- When the variance is not known we have to calculate $S^2 = \sum_{i=1}^n (X_i - \bar{X})^2 / (n - 1)$ and S^2/n is then the variance of the sampling distribution

95% confidence interval ($z_{\zeta/2} = 2.571$ following a Student T-distribution as $n < 30$):

$$[\bar{X} - \frac{z_{\zeta/2}S}{\sqrt{n}}, \bar{X} + \frac{z_{\zeta/2}S}{\sqrt{n}}] = [4.6577, 5.1139]$$

Sampling

- Interval estimators/Confidence intervals: Example 2
 - Suppose that the city of Ghent interview respondents via phone calls to conduct the survey. The duration of telephone calls is normally distributed (not required) with population mean $\mu=180$ seconds and standard deviation $\sigma = 30$ seconds. Ten calls are chosen at random and sampled.
 - Since the duration of a call is normally distributed, the 10 random calls $X_1, X_2, \dots, X_{10}$ are also normally distributed and $\bar{X} = \frac{1}{n}(X_1 + X_2 + \dots + X_{10}) \sim N(180, 90)$
 - Calculate the probability that $\bar{X}$ is in error by more than 15 seconds

$$\begin{aligned}\text{Prob}\{|\bar{X} - \mu| > 15\} &= 1 - \text{Prob}\{|\bar{X} - \mu| \leq 15\} \\ &= 1 - \text{Prob}\{-15 \leq \bar{X} - \mu \leq 15\} \\ &= 1 - \text{Prob}\left\{\frac{-15}{\sqrt{30^2/10}} \leq \frac{\bar{X}-\mu}{\sigma} \leq \frac{15}{\sqrt{30^2/10}}\right\} \\ &= 1 - \text{Prob}\{-1.5811 \leq Z \leq 1.5811\} = 2(1 - \text{Prob}\{0 \leq Z \leq 1.5811\}) = 0.1142\end{aligned}$$

Sampling

- Interval estimators/Confidence intervals: Sample size determination

- Approach

Step 1. Choose a value of n (where n is at least 30) and generate n samples $X_1, X_2, \dots, X_n$

Step 2. Compute the sample mean $\bar{X}$ and the sample variance S^2

Step 3. If $S/\sqrt{n}$ is sufficiently small, then stop. Otherwise generate additional samples and go back to step 2.

- Example

- Consider the following sample of 16 numbers:

11.6; 13.5; 9.1; 13.9; 12.1; 9.6; 10.7; 9.5; 12.7; 10.8; 8.1; 11.4; 12.4; 11.2; 9.8; 6.1

- The sample mean $\bar{X} = 10.7812$ and the sample variance $S^2 = 62.3244/15 = 4.155$.
- The standard deviation of the sampling distribution $\bar{X} = S/\sqrt{n} = 2.0384/4 = 0.5096$
- At the 95% confidence interval, we can assert that the sample mean of 10.7812 does not differ from the true mean more than $2.131 \times 0.5096 = 1.086$ ($\Rightarrow$ use of Student t-distribution since $n = 16 < 30$)

Sampling

- Interval estimators/Confidence intervals: Sample size determination

- Example

- To estimate the number of samples needed for the standard deviation of the estimator to be less than 0.25, we need the value of n such that $S/\sqrt{n} < 0.25$. Given that $S = 2.0384$ the value of n must be greater than $66.4793 = 67$.
- When $n = 67$, the maximal margin of error ϵ (i.e. $Z_{\zeta/2} \frac{S}{\sqrt{n}} \leq \epsilon$) with $1-\zeta$ is the required accuracy (e.g. 95% confidence interval).

$$\max \epsilon = Z_{2.5\%} \frac{S}{\sqrt{n}} = 1.96 \times \frac{2.0384}{\sqrt{67}} = 0.488$$

- The required sample size is calculated by the following formula

$$n \geq \left(\frac{Z_{\zeta/2} \times S}{\epsilon} \right)^2$$

(The sample size should be a whole number, i.e. you have to round up)

Sampling

- Dependence between observations

- Example

- If the patients at a medical facility have to wait too long to see a doctor, they leave the facility and go elsewhere. A simulation experiment may lead to some insights in the waiting time of the patients (X_i = the waiting time of the i^{th} patient). However, the following problems arise:
 - Confidence intervals are meaningless since the observations X_i and X_{i+1} are correlated (if patient i has a long waiting time, patient $i+1$ is very likely to have a long waiting time as well)
 - The initial state at the start of the simulation impact the results (e.g. In the beginning the facility is idle and early scheduled patients wait little)

Sampling

- Dependence between observations

- Transient vs steady state simulation

- Transient or terminating simulation

- The simulation period is well defined and relatively short
- The initial state of the system plays an important part in its evolution

⇒ Independent samples are obtained by running the simulation many times, using exactly the same initial conditions and terminating at the same time instant or stopping event (*independent replications* with different seeds for the random number generation)

- Suppose

N the number of simulation runs

n the number of samples for each simulation run

X_{ik} the k^{th} sample during the i^{th} simulation run

$\bar{X}_i$ the point estimator of the sample mean for simulation run i $\bar{X}_i = \frac{1}{n} \sum_{k=1}^n X_{ik}$

$\bar{\bar{X}}$ the point estimator of the sampling distribution $\bar{\bar{X}} = \frac{1}{N} \sum_{i=1}^N \bar{X}_i$

⇒ Since the point estimators $\bar{X}_i$ are independent we are able to calculate the sample variance and confidence intervals

Sampling

- Dependence between observations

- Transient vs steady state simulation

- Steady state or nonterminating simulation

- No fixed beginning instant nor an ending instant
- Independent from the initial state

⇒ The simulation results prior to the steady state, i.e. the transient period, should be discarded and monitor the values until they appear to settle in some sort of regular pattern. However, it is difficult to calculate the end of the transient period

- Suppose

n the number of samples in a simulation run

k the number of samples impacted by the initial state

X_i the i^{th} sample in the simulation run

$\bar{X}$ the point estimator of the sample mean of the simulation run

$$\bar{X} = \frac{1}{(n-k)} \sum_{i=k+1}^n X_i$$

Sampling

- Dependence between observations

- Transient vs steady state simulation

- Steady state or nonterminating simulation

- Since subsequent observations X_i and X_{i+1} may be correlated, we obtain independent samples via many independent replications by and running the simulation many times (Drawback: long runtimes as transient period must be eliminated)
 - Use the batch means method: Perform one simulation, remove the transient phase and divide the remainder of the simulation into a set of N partial simulations or batches (with a length to serve n customers or with a fixed time duration). The length of a batch is equal to $5L$ assuming that the correlation between X_i and X_{i+L} is negligible.
 - The problem if eliminating transient portion of a steady-state simulation run is non-existent in the case of a *regenerative process*
 - A regenerative process is one in which the system regenerates itself from time to time, essentially beginning from scratch
 - The length of time between consecutive regeneration instants are called regeneration cycles and random variables in one regeneration cycle are independent from other regeneration cycles.

Monte Carlo Integration

- The relationship between Monte Carlo Simulation and Monte Carlo Integration

- Monte Carlo Simulation

- The imitation of a real-world process by using random number generation and probability distributions to randomly generate events that occur in the system. Running the simulation model, results in a statistical observation of the performance of the system

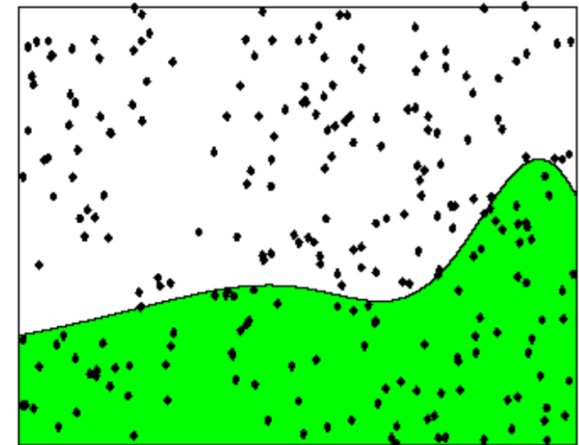
- Monte Carlo Integration

- Monte Carlo integration uses random sampling of a function to numerically compute an estimate of its integral
- Monte Carlo Integration calculates the area 'under the curve' or the first moment (the mean) of a random variable X
- This can be computationally expensive for some functions that do not have an analytical closed-form expression. Monte Carlo Integration can deliver a reasonably good result at a cheaper computational cost. The larger the number of samples or simulation runs the better the estimate (Law of Large Numbers/Central Limit Distribution)

Monte Carlo Integration

- Basic idea – Method I

- Draw random number in the x - y plane (dots in the graph)
- Integral of function f is approximately given by the total area times the fraction of points that fall under the curve $f(x)$
- The larger the number of points the more accurate the evaluation of this area
- Method is only competitive for complicated and/or multi-dimensional functions
- Cf. Excel file 'Session 2 - Monte Carlo Integration.xls'



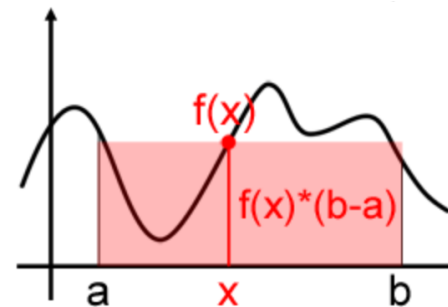
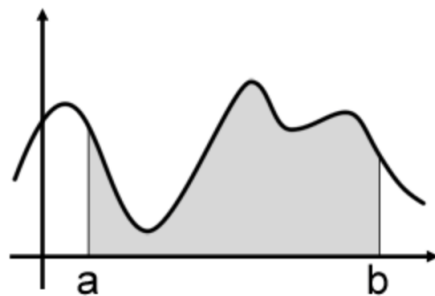
Monte Carlo Integration

- Basic idea – Method 2

➤ Suppose we want to integrate a one-dimensional function $f(x)$

$$F(x) = \int_a^b f(x)dx$$

- The integral of a function $f(x)$ can be interpreted as calculating the area under the curve (*left figure*)
- Pick a random value for x , $x \in [a, b]$, and evaluate the function $f(x)$ at x . This is a very crude approximation of the area under the curve: The area under the curve = $f(x) \times (b-a)$ (*right figure*)

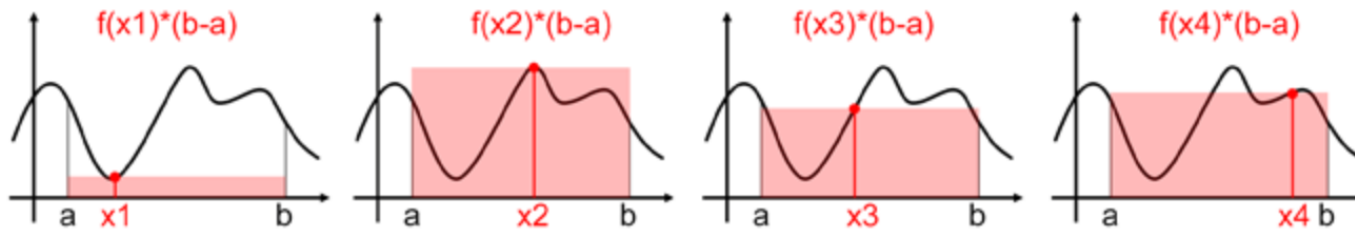


Monte Carlo Integration

- Basic idea – Method 2

- Multiple samples are required for an accurate estimation

- Picking one random value leads to an under- or overestimation of the area
- Pick multiple random values for x , $x \in [a, b]$, and evaluate the function $f(x)$ at x



- Averaging the results over the different samples leads to a better estimation of the integral. This approximation converges to the integral result as the sample size increases or $E[\bar{X}] = \mu$

$$\bar{X} = (b - a) \frac{1}{n} \sum_{i=1}^n f(X_i)$$

with $\bar{X}$ the sample mean value estimating the area under the curve

$$X_i = a + u_i(b - a) \text{ and } u_i \text{ is a uniform random number between 0 and 1}$$

Monte Carlo Integration

- The formulas

- Calculate the expected value of the integral

$$\int_{\Omega} g(x) dx = \int_{\Omega} \frac{g(x)}{f(x)} f(x) dx = \int_{\Omega} h(x) f(x) dx = E_f[h(X)]$$

with $x \sim f(x)$ is the sample distribution and for any f such that

1. f is defined for all $x \in \Omega$

2. f is always nonnegative: $f(x) \geq 0, \forall x \in \Omega$

3. f integrates to 1: $\int_{x \in \Omega} f(x) dx = 1$

$h(x)$ is the target distribution

- The integration corresponds to calculating a mean value. Use the sample mean as point estimator:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n h(X_i)$$

where X_i is a sampling point from the function $f(x)$ or $X_i \sim f(x)$

Monte Carlo Integration

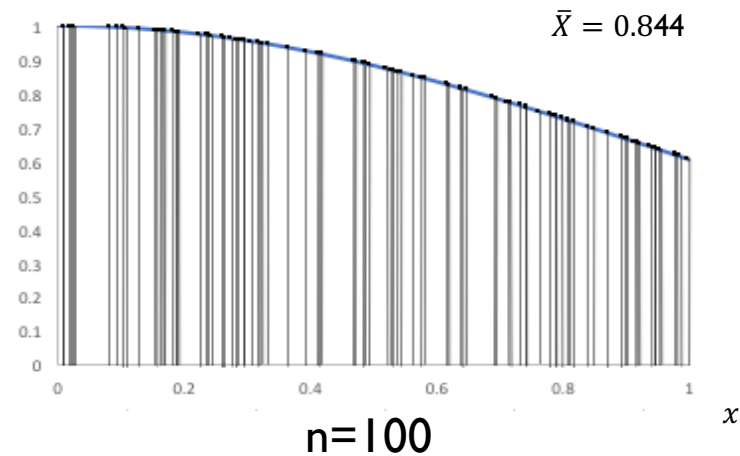
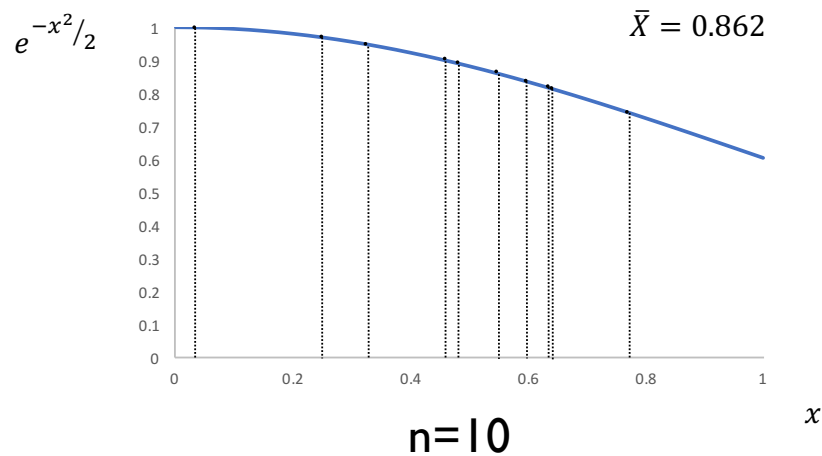
- Example I: Definite integral

➤ Calculate the expected value of the following integral

$$\int_0^1 e^{-x^2/2} dx$$

➤ Then set

- $h(x) = e^{-x^2/2}$
- $f(x) = 1$, i.e. $x \sim \text{Unif}(0,1)$



Monte Carlo Integration

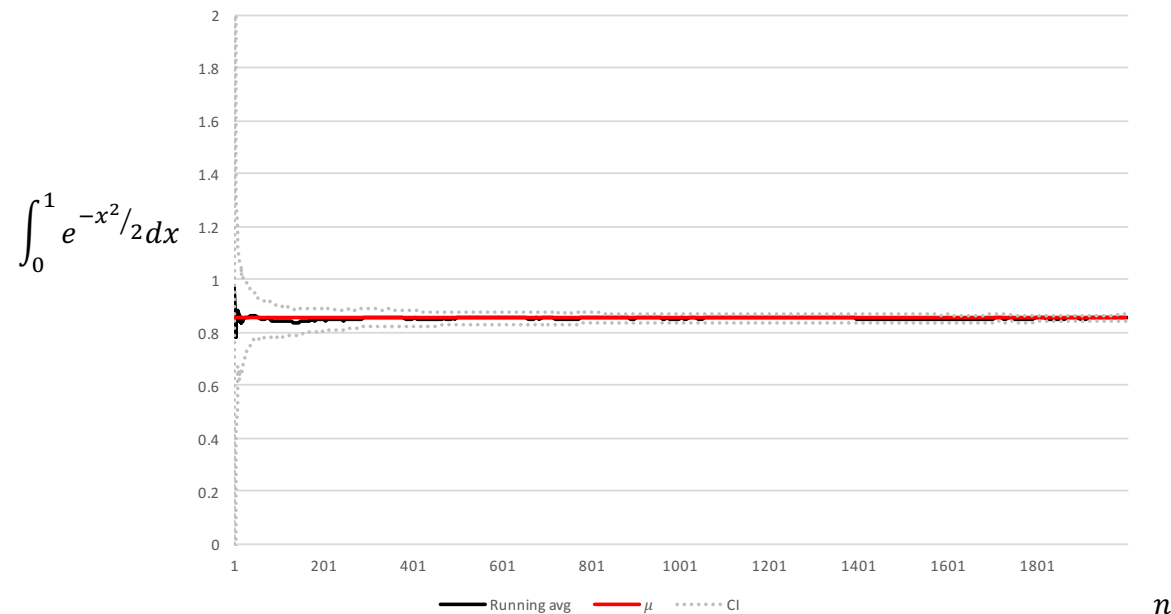
- Example I: Definite integral

➤ Calculate the expected value of the following integral

$$\int_0^1 e^{-x^2/2} dx$$

➤ Then set

- $h(x) = e^{-x^2/2}$
- $f(x) = 1$, i.e. $x \sim \text{Unif}(0,1)$



Monte Carlo Integration

- Example 2: Infinite bounds

➤ Calculate the expected value of the following integral

$$E_{N(0,1)}[X] = \int_{-\infty}^{+\infty} x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$$

➤ Then set

- $h(x) = x$
- $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} = \phi(x)$, i.e. $x \sim N(0,1)$



Monte Carlo Integration

- Example 3: Outpatient Scheduling
 - Monte Carlo method aims at evaluating complex integrals
 - Most often we do not specify the integral representation
 - This example shows how it relates to simulation
 - Session in outpatient clinic
 - Patients arrive at appointed times in the clinic (no lateness)
 - Patients have a stochastic consultation time
 - Trade-off:
 - Consultation ends prior to the arrival of the next patient → physician must wait
 - The next patient arrives when the physician is busy → patient must wait

Monte Carlo Integration

- Example 3: Outpatient Scheduling

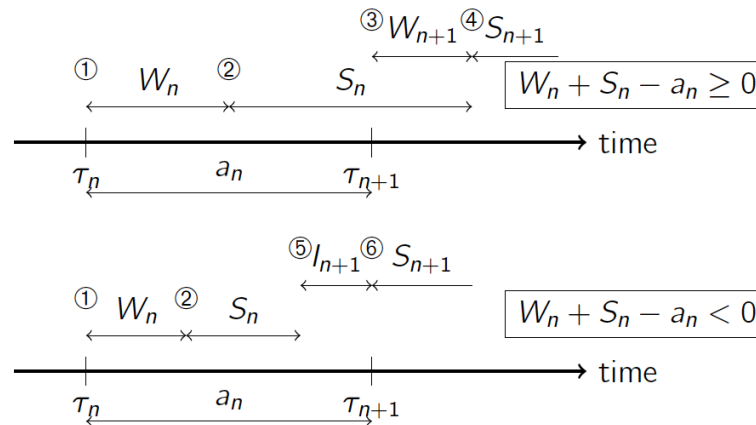
- Notation

t_{max}	length of the session
τ_n	appointment time of the n^{th} patient ($0 \leq \tau_1 \leq \tau_2 \leq \dots \leq \tau_N \leq t_{max}$)
d	arrival time of the physician ($0 \leq d \leq t_{max}$)
S_n	consultation time of the n^{th} patient
a_n	interarrival time, i.e. $\tau_{n+1} - \tau_n$

Monte Carlo Integration

- Example 3: Outpatient Scheduling

- Relation between consecutive waiting times



1. Patient n arrives has to wait W_n minutes
2. Patient n enters the physician's office
3. Patient $n+1$ arrives while patient n is being seen

4. patient n leaves and patient $n+1$ enters the physician's office
5. patient n leaves and the physician waits for the next patient
6. patient $n+1$ arrives and does not have to wait

Monte Carlo Integration

- Example 3: Outpatient Scheduling

➤ Formula for the n^{th} waiting time

$$\begin{aligned} W_n &= \max(W_{n-1} + S_{n-1} - a_{n-1}, 0) \\ &= \max(\max(W_{n-2} + S_{n-2} - a_{n-2}, 0) + S_{n-1} - a_{n-1}, 0) \\ &= \max(W_{n-2} + S_{n-2} - a_{n-2} + S_{n-1} - a_{n-1}, S_{n-1} - a_{n-1}, 0) \\ &= \dots \\ &= \max(W_1 + \sum_{j=1}^{n-1} S_j - a_j, \sum_{j=2}^{n-1} S_j - a_j, \dots, \sum_{j=n-1}^{n-1} S_j - a_j, 0) \\ &= \max(\max(d - \tau_1, 0) + \sum_{j=1}^{n-1} S_j - a_j, \sum_{j=2}^{n-1} S_j - a_j, \dots, \sum_{j=n-1}^{n-1} S_j - a_j, 0) \\ &= G_n(S_1, S_2, \dots, S_{n-1}) \end{aligned}$$

Monte Carlo Integration

- Example 3: Outpatient Scheduling

- Mean waiting time

$$E[W_n] = \int_0^\infty \int_0^\infty \dots \int_0^\infty G_n(s_1, s_2, \dots, s_{n-1}) dS_1(s_1) dS_2(s_2) \dots dS_{n-1}(s_{n-1})$$

=> This integral is not easy to calculate

- Use Monte Carlo estimators

- Sample the consultation times K times

$$S^{(\ell)} = [S_1^{(\ell)}, S_2^{(\ell)}, \dots, S_N^{(\ell)}] \quad \ell = 1, \dots, K$$

- Use the formulas to calculate the waiting times

$$W_{n+1}^{(\ell)} = \max(W_n^{(\ell)} + S_n^{(\ell)} - a_n, 0) \quad \Rightarrow W^{(\ell)} = [W_1^{(\ell)}, W_2^{(\ell)}, \dots, W_N^{(\ell)}]$$

Monte Carlo Integration

- Example 3: Outpatient Scheduling

- Use Monte Carlo estimators

- Calculate the sample average

$$\bar{X} = \frac{1}{K} \sum_{\ell=1}^K W^{(\ell)}$$

Variance reduction methods

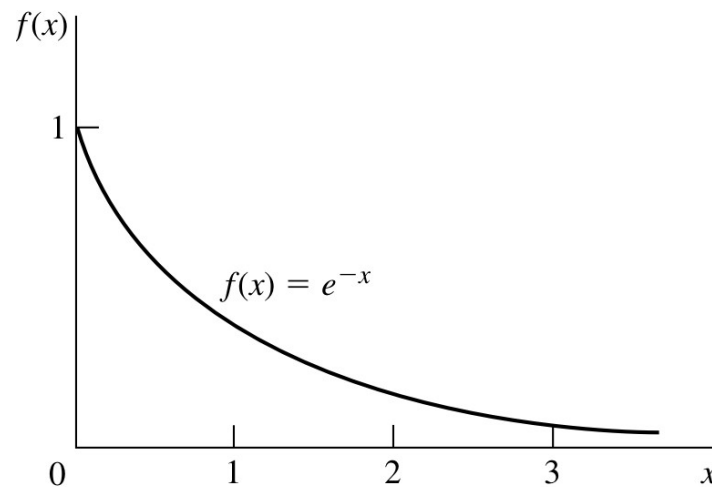
- Error for Monte Carlo Methods $\sim \sqrt{\text{var}(h(X))/n}$
 - To reduce this error we may increase the number of samples (very slow)
 - Limit the required number of simulated samples via a reduction in the variance of parameter estimators. Some techniques for increasing the precision (i.e., decreasing the variance) of sample estimators:
 - Stratified sampling
 - Antithetic variables
 - Control variables
 - Importance sampling
 - Conditioning

Variance reduction methods

- Problem: Crude random sampling may not cover certain (critical) portions of the distribution.
- Example
 - Exponential distribution with a parameter $\lambda = 1$

Probability density function: $f(x) = e^{-x}$

Cumulative distribution function: $F(x) = 1 - e^{-x}$



Variance reduction methods

- Example

- Suppose that mean is not known and we want to estimate this mean using simulation

- Crude Monte Carlo technique: Generate some random observations and average these estimations

<i>l</i>	Random Number* r_i	Random Observation $x_i = -\ln(1 - r_i)$
1	0.495	0.684
2	0.335	0.408
3	0.791	1.568
4	0.469	0.633
5	0.279	0.328
6	0.698	1.199
7	0.013	0.014
8	0.761	1.433
9	0.290	0.343
10	0.693	1.183
		Total = 7.793
		Estimate of mean = 0.779

Known mean $\mu = 1/\lambda = 1$
Known st. dev. $\sigma = 1/\lambda^2 = 1$

St dev sampling distribution
= $\sigma/\sqrt{n} = 1/\sqrt{10}$



Poor accuracy

Variance reduction methods

- Stratified sampling
 - The distribution is divided into portions (called *strata*) and each portion is sampled individually, with disproportionately heavy sampling of more critical strata.

Stratum	Portion of Distribution	Stratum Random No.	Sample Size	Sampling Weight
1	$0 \leq F(x) \leq 0.64$	$r'_i = 0 + 0.64r_i$	4	$w_i = \frac{4/10}{0.64} = \frac{5}{8}$
2	$0.64 \leq F(x) \leq 0.96$	$r'_i = 0.64 + 0.32r_i$	4	$w_i = \frac{4/10}{0.32} = \frac{5}{4}$
3	$0.96 \leq F(x) \leq 1$	$r'_i = 0.96 + 0.04r_i$	2	$w_i = \frac{2/10}{0.04} = 5$

Variance reduction methods

- Stratified sampling

Stratum	l	Random Number r_i	Stratum Random No. r'_i	Stratum Random Observation $x'_i = -\ln(1 - r'_i)$	Sampling Weight w_i	x'_i / w_i
1	1	0.495	0.317	0.381	$\frac{5}{8}$	0.610
	2	0.335	0.215	0.242	$\frac{5}{8}$	0.387
	3	0.791	0.507	0.707	$\frac{5}{8}$	1.131
	4	0.469	0.300	0.357	$\frac{5}{8}$	0.571
2	5	0.279	0.729	1.306	$\frac{5}{4}$	1.045
	6	0.698	0.864	1.995	$\frac{5}{4}$	1.596
	7	0.013	0.644	1.033	$\frac{5}{4}$	0.826
	8	0.761	0.884	2.154	$\frac{5}{4}$	1.723
3	9	0.290	0.9716	3.561	5	0.712
	10	0.693	0.9877	4.398	5	0.880

Total = 9.481
Estimate of mean = 0.948

Variance reduction methods

- Antithetic variables or complementary random numbers

➤ Suppose two variables X_1 and X_2 with mean μ and variance σ^2

- $E\left[\frac{X_1+X_2}{2}\right] = \frac{E[X_1]+E[X_2]}{2} = E[X]$

- $Var\left[\frac{X_1+X_2}{2}\right] = \frac{1}{4}[Var[X_1] + Var[X_2] + 2Cov(X_1, X_2)] = \frac{Var[X]+Cov(X_1, X_2)}{2}$ with $Cov(X_1, X_2) = E[X_1X_2] - \mu^2$

- The antithetic method attempts to introduce negative correlation into the samples which decreases the variance

- Consider $X_1 = F_X^{-1}(U)$ whereby X_1 is a function of k uniformly distributed random numbers $u_1, u_2, \dots, u_k$

$$X_2 = F_X^{-1}(V) = F_X^{-1}(1 - U) \text{ whereby } X_2 \text{ is a function of } k \text{ uniformly distributed random numbers } v_1, v_2, \dots, v_k \text{ and } u_i = 1 - v_i$$

⇒ As a result, the variance is reduced (because of the negative correlation between X_1 and X_2 and we should generate less random numbers)

⇒ $E[X_1] = E[X_2] = \frac{1}{2}E[X_1 + X_2]$ or $E[h(X_1)] = E[h(X_2)] = \frac{1}{2}E[h(X_1) + h(X_2)]$ (i.e. the average can be used to estimate the performance measure)

⇒ For monotone function h , there is a negative correlation between $h(X_1)$ and $h(X_2)$ if X_1 and X_2 are negatively correlated

⇒ We have twice as many samples and twice as many evaluations of X but not twice as many random numbers

Variance reduction methods

- Antithetic variables or complementary random numbers
 - The method requires that X_1 and X_2 are negatively correlated, which is ensured if X_1 is generated via the inversion method. If X_1 is generated via the acceptance/rejection method the negative correlation is not certain.
 - Intuitive meaning
 - The “luck of the draw” on uniform random numbers may cause the average of the resulting random observations to be on one side of the true mean.
 - The complements of those uniform random numbers then tend to yield the opposite result.
 - Therefore, using both and then calculating the combined sample average should provide a more precise estimator.

Variance reduction methods

- Antithetic variables or complementary random numbers

➤ Example

I	Random Number r_i	Random Observation $x_i = -\ln(1 - r_i)$	Complementary Random Number $r'_i = 1 - r_i$	Random Observation $x'_i = -\ln(1 - r'_i)$
1	0.495	0.684	0.505	0.702
2	0.335	0.408	0.665	1.092
3	0.791	1.568	0.209	0.234
4	0.469	0.633	0.531	0.756
5	0.279	0.328	0.721	1.275
6	0.698	1.199	0.302	0.359
7	0.013	0.014	0.987	4.305
8	0.761	1.433	0.239	0.272
9	0.290	0.343	0.710	1.236
10	0.693	1.183	0.307	0.366

Total = 7.793

Total = 10.597

$$\text{Estimate of mean} = \frac{1}{2}(0.779 + 1.060) = 0.920$$

Variance reduction methods

- Antithetic variables or complementary random numbers

- Example (cf. Example 1 – Monte Carlo Integration)

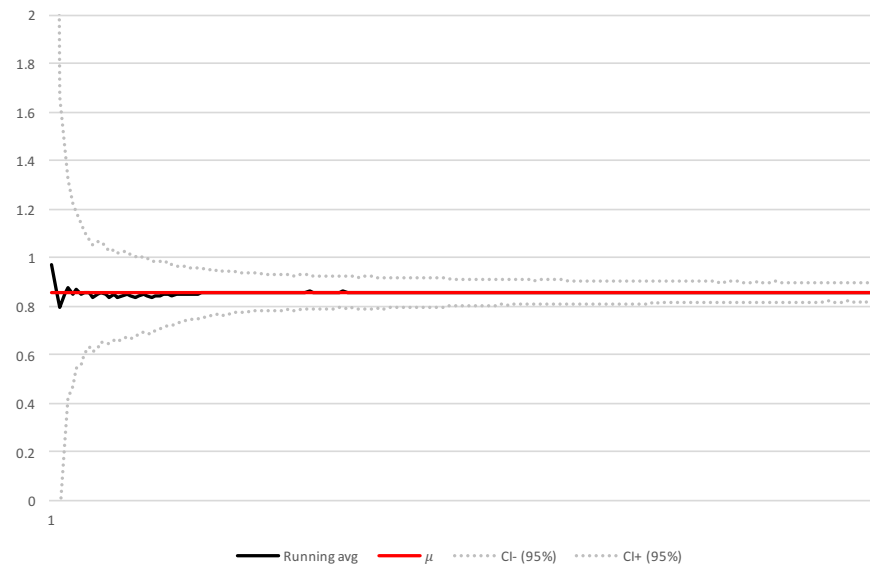
- Without Antithetic method ($n = 200$): $\bar{X} = 0.84517$; $S = 0.30036$; 95% CI = $[0.80355; 0.8868]$

$$\text{Cov}(\text{First } 100, \text{Second } 100) = 0.0019$$

- With Antithetic method ($n = 100 + 100$): $\bar{X} = 0.85740$; $S = 0.27969$; 95% CI = $[0.81864; 0.89616]$

$$\text{Cov}(\text{First } 100, \text{Second } 100) = -0.0784$$

- True mean $\mu = 0.8556$



Variance reduction methods

- Control variates

- The principle

- Obtain a second set of identically distributed samples $Y_i, i = 1, 2, \dots, n$ such that Y_i is correlated with X_i and for which $E[Y_i] = v$ can be calculated analytically and is known
- The correlation may be positive or negative but the stronger the better. The value $\bar{Y} = \sum_{i=1}^n Y_i / n$ can be used to modify $\bar{X}$ depending upon whether $\bar{Y}$ is greater than or less than v
- Suppose that $\bar{X}$ and $\bar{Y}$ are positively correlated, so that a large value of $\bar{Y}$ implies that $\bar{X}$ is large and a small value of $\bar{Y}$ implies that $\bar{X}$ is small
 - If $\bar{Y} > v$, we can expect that $\bar{X}$ is larger than its mean μ and we need to adjust $\bar{X}$ downwards
 - If $\bar{Y} < v$, we can expect that $\bar{X}$ is smaller than its mean μ and we need to adjust $\bar{X}$ upwards(vice versa for a negative correlation)
- Since $\bar{Y}$ controls the value of $\bar{X}$, we use the so-called method of control variables

Variance reduction methods

- Control variates

- The formulas

- If $\bar{X}$ is an unbiased estimator of X , $\bar{X}_c = \bar{X} - c(\bar{Y} - v)$ is also an unbiased estimator

- $\bar{X}_c = \frac{1}{n} \sum_{i=1}^n (X_i - c(Y_i - v))$
- $E[\bar{X}_c] = E[\bar{X}] - c(E[\bar{Y}] - E[v]) = E[\bar{X}] - c(v - v) = E[\bar{X}] = \mu$
- $\text{Var}[\bar{X}_c] = \text{Var}[\bar{X}] + c^2 \text{Var}[\bar{Y}] - 2c \text{Cov}(\bar{X}, \bar{Y})$ and this variance is minimised if $c = \text{Cov}(\bar{X}, \bar{Y}) / \text{Var}[\bar{Y}]$

(note that the function is convex $\frac{d^2 f(c)}{dc^2} > 0$ and a minimum is reached where $\frac{df(c)}{dc} = 0$)

- As a result $\frac{\text{Var}[\bar{X}_c]}{\text{Var}[\bar{X}]} = 1 - [\text{Corr}(\bar{X}, \bar{Y})]^2$ where $\text{Corr}(\bar{X}, \bar{Y})$ is the correlation between $\bar{X}$ and $\bar{Y}$. This formula shows that when $\bar{X}$ and $\bar{Y}$ are correlated the variance of $\bar{X}_c$ is smaller than the variance of $\bar{X}$.
- Unfortunately the value of c cannot be calculated in an analytical manner because the $\text{Cov}(\bar{X}, \bar{Y})$ is not known

$$\widehat{\text{Cov}}(\bar{X}, \bar{Y}) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n-1} \text{ and } \widehat{\text{Var}}(\bar{Y}) = \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n-1} \text{ and } \hat{c} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$$

Variance reduction methods

- Control variates

- Example

- Consider an M/M/1 queue in which the mean interarrival time ($1/\lambda$) amounts 5 minutes and the mean service time ($1/\mu$) is 4 minutes. We conduct ten simulation runs. Each run starts with an idle system and terminates when 200 customers are served.
 - For each simulation run i , we report the average amount of time spent by customers waiting in the queue (X_i) and the mean service time (Y_i) which acts as a control variable.

i	X_i	Y_i
1	18.79	4.15
2	17.97	3.82
3	18.54	4.09
4	9.58	3.53
5	14.51	4.01
6	11.61	3.90
7	9.87	3.78
8	16.66	3.94
9	27.09	5.24
10	22.92	4.97

Variance reduction methods

- Control variates

- Example

- Based on these observations: $\bar{X} = 16.754$ and $\bar{Y} = 4.143$

- We know: $E[\bar{Y}] = 4$ and $E[\bar{X}] = \frac{\lambda/\mu}{(\mu-\lambda)} = 16$

=> Both $\bar{X}$ and $\bar{Y}$ overestimate their respective means (which supports the intuition that both are positively correlated)

- The variances and covariances are calculated as follows:

$$S_X^2 = \frac{1}{9} \sum_{i=1}^9 (X_i - \bar{X})^2 = 31.5586 \quad S_Y^2 = \frac{1}{9} \sum_{i=1}^9 (Y_i - \bar{Y})^2 = 0.291 \quad S_{X,Y} = \frac{1}{9} \sum_{i=1}^9 (X_i - \bar{X})(Y_i - \bar{Y}) = 2.7143$$

- The correlation between X and Y: $\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{2.7143}{\sqrt{31.5586} \sqrt{0.2911}} = 0.8955$ (positive!)

- The optimum value for c: $c_{opt} = \frac{S(X,Y)}{S_Y^2} = 9.3243$

- The final approximation: $\bar{X}_c = \bar{X} - c(\bar{Y} - v) = 16.754 - 9.3243(4.143 - 4) = 15.4207$ (closer to the true mean)

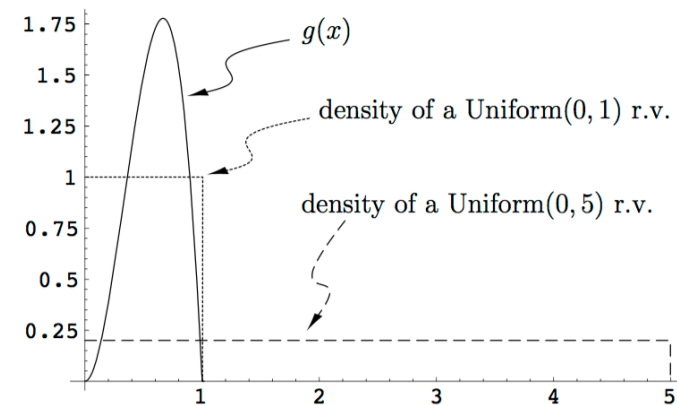
- The variance: $S_c^2 = \text{Var}[\bar{X}] + c^2 \text{Var}[\bar{Y}] - 2c \text{Cov}(\bar{X}, \bar{Y}) = 31.5586 + 9.3243^2 \times 0.2911 - 2 \times 9.3243 \times 2.7143 = 6.2505$

Variance reduction methods

- Importance sampling

➤ Calculate the expected value of the following integral

$$\int_0^1 g(x)dx = \int_0^1 \frac{g(x)}{f(x)} f(x)dx = \int_0^1 h(x)f(x)dx = E_f[h(X)]$$



Note that $g(x) = 0$ for $x < 0$ and $x > 1$

- $x \sim f(x) = \text{Unif}(0,1)$
 - Monte Carlo Estimate: $\bar{X} = \frac{1}{n} \sum_{i=1}^n h(X_i)$

Variance reduction methods

- Importance sampling

- The choice of the sampling distribution may reduce the required number of samples

- $x \sim f(x) = \text{Unif}(0,1)$ $\Rightarrow$ Monte Carlo Estimate: $\bar{X} = \frac{1}{n} \sum_{i=1}^n h(X_i)$

- (this method will work reasonably well)

- $x \sim f(x) = \text{Unif}(0,5)$ $\Rightarrow$ Monte Carlo Estimate: $\bar{X} = (b - a) \frac{1}{n} \sum_{i=1}^n h(X_i) = \frac{5}{n} \sum_{i=1}^n h(X_i)$

- This method makes no sense since 80% of the realised random numbers X_i will tell you nothing substantial about the integral since $g(x) = 0$ for $1 < x < 5$.

$\Rightarrow$ The sampling distribution $f(x)$ should only take values in the considered interval relevant for the integral

Variance reduction methods

- Importance sampling

➤ Consider the following mean of the random variable X

$$\int_{\Omega} g(x) dx = \int_{\Omega} \frac{g(x)}{f(x)} f(x) dx = \int_{\Omega} h(x) f(x) dx = E[h(X)]$$

where $f(x)$ is a probability density function defined over the range Ω of values x and with $h(x) = \frac{g(x)}{f(x)}$ and $X \sim f(x)$

➤ Question: How should we choose $f(x)$ for minimal variance?

- We would like to minimise $Var(h(X)) = E[(h(X) - \mu)^2] = \int_{\Omega} [\frac{g(x)}{f(x)} - \mu]^2 f(x) dx$
- If $\forall x \in \Omega: g(x) \geq 0$ and μ is known then the density $f_{opt}(x) = \frac{g(x)}{\mu}$ makes the variance 0
- However, μ is not known but we have to use a density that resembles $g(x)$ as good as possible

Variance reduction methods

- Importance sampling

- A good sampling distribution $f(x)$ should have the following properties

- $f(x) > 0$ whenever $h(x) \neq 0$
 - $f(x)$ should be close to being proportional to $|h(x)|$
 - It should be easy to simulate values from $f(x)$
 - It should be easy to compute the density $f(x)$ for any value x that you might realise

Reference:

Anderson, E.C., 1999, Monte Carlo Methods and Importance Sampling, Statistical Genetics, 1-8.